Patrick Call

AI Specialist -> 2 years out of school, electrical and swe, queen's, in toronto next week, wnat to meet with him

What he would do if I were him

Project or two, I can help out with over the summer, doesn't know

Value based investing approach, kind of like warren buffet, long only, equity focused

Only invest in the funds, -> only driven by own personal gain, personal wealth first, then clients get onto

Return Centric, Agnostic to volatility, index plus game, don't want to play it

Volatility is a function of crowd behaviour

Vision to start a different fund -> smart alpha, 5 or 6 different wy, triangultae, 2 or 3 together and act on them

Old school -> 30 years in the business

Machines, computers, algorithms, AI -> how to create an opportunity

Think of the new world -> AI, deeper learning, think about it, done some work on volatility, some statistics behind

Brought something to table, testing for alpha generation on different data, uncorrelation, p-statistics

What computer systems could you bring, different ways to think about the market

Aidan -> new hire, landing in toronto on the 3rd, 4th and 5th, 6th, 9th, 13th

AI conference, 9th, video call, probably not on the call

Very smart -> specializes in AI and Machine learning, statsitsic and finance theory that is where his weaknesses are

Haven't seen an opportunity like this -> business processes

New fund -> Equities easiest to do, unconstrained at this point, commodities as well, further out from equities -> more complex it gets

Equities is the preference, multi asset class also fine

Value manager -> taking the expected future cashflows, and the difference in the price, then keep midn open to macros, open to being contrarian, no great depression (policy mistakes), learned those lessons, not goes into that

Macros affect volatility, returns and drawdowns, and live with that

Portfolio excels at this -> world expresses this, 5 -6 years, growth environment, macros -> declining interest rates

How do you take advantage of growth environment or value environment

Value is over, or you buy growth stocks -> value drift, abandoned the drift,

Don’t believe too much in growth, fleeting

Astounded

Created a new fund, one of the -> long value or long growth, bucket of growth or bucket of value, ratio together -> do you wnat to be in value or growth, see from intercept growth, up or down, you can tell that , line breaks that moving average

What is the right bucket, what is the right moving average, how much

Segeratgate the market into different components, and also defensives,

Portfolio rebalancing, based

Value-based pools, Growth ETF, Defensive ETF

Otehr sectors as well, which you can classify as growth or defensive

Can play this game, or equal weighting, large cap or small cap

And switching the factors

Or you can do it by basket -> long growth or short value

Purely opportunistic

You just care about short term, your profits

Launch fund -> long term goal

Aidan -> coming on board to help with processes in AI

7 to 8 maybe 10 different models, statistical relationship

Try to triangulate and look for patterns, doing this in terms of in his head

Looks at sell side analysts etc

Drillers, trading at 3X EBITDA, no CAPEX, in 3 years you will get entire company for free, many things need to go wrong

Why not mean reversion

Potential Suggestion -> Patrick Agent to assist your analysts, insiders, breaks out technically, bottom triangle, reaching new highs, very distressed

7 or 8 silos -> lot more what we can do

Patrick LLM

It maybe what the project is

Gun being fired

ADBE -> Chat two to three

4 to 5, 5-> 5.2

How good can AI get?

Agentic AI is big

ChatGPT -> knows how I think

Believes AI will be smart enough to build your own stuff and tailor it

Your moat -> model needs to be spectacular

Daughter is in advertising

Software world -> Biggest risk , is the sunk cost that the barriers to entry is coming down

It can create code the way you think, way more powerful than the SaaS model

3D printers, revolutionalize a lot of things, given how much money into AI and data centres, doesn't think they are going to get Hyperscalers

AI will break down many SaaS models -> mostly many are going to be affected

Contrarian skills are very important, but macros shift valuation rewards; what you think may be the opportunities, may not have been the opportunties

Dream100 -> opportunity

Dynamic

Investment managers-> IGM, and AGI, private equity is going down -> private markets -> 30% to 40% in stocks, bloody mess

Run public funds, -> very different in terms -> money moves away from private debt and private equities, liquidity risk

Discussing with Aidan, on this monday,

Openness, and also, many key holes, no crystal ball, your job is tofigure out many as many key holes, have conviction

Nothing immediate

Aidan Lunch

Notes from lunch

Ate Chicken Milanese -> very delicious, should have asked how is the food, how is the restaurant, $27 for the food, with tax about $30

Company paid for lunch

Aidan is their son, Patrick's son

Patrick AI model -> building an AI to help his existing decision making

Spoke about my story from math -> coding -> accounting -> finance -> quant/fintech

Spoke about his motivation to do something separate then parents

Talked about product market fit and the technical founders being with product obsession rather than customer obsession

Explained the modeling bit to him -> Prop firm monte carlo simulation modelling

Mentioned HMM modeling to him

Picthed sentiment analyzer along with the future scenarios

Explained it using US Iran -> Oil prices changing, mentioned specific numbers

Talked abotu how Andrea documents more than Patrick

Need to get historical data to feed the model

This is an experimental project, lot of freedom, self agency, potential to build something huge

Masters student in Stats possibly working with

Gave him the industry idea -> different industries understanding

I also explained to him how I found them -> AI workflow, he was super impressed by this as well

Want to make something align

Gave him idea of the transcribing the investment memos, comparing the model output ot his output and then doing it again and again

Self Agency in this project -> starting a fund, very entrepreeurial

They want me -> Next steps, he will think about the exact things I will work on in this project, and also wants me to think about examctly what I would do on the day to day, as well

Said to send him the ideas, that you can work on,

Basically my job is to find new alpha -> quant research internship, statistical modeling, finding new alpha for an equities fund, learning the entire process

Now, it will be important to coffee chat the two quants from the UTEFA thingy, to potentially grab a coffe with them, ask them how they would approach this

Learn, the python required for backtesting, the probability and stats required, enough lin alg, ECO220 definitely do it in the summer, and learn the time series modeling enough deeply that I can do all of these things

And go deep into the financial intuition that your strategies are going to be based on, basically founding a fund, it is already funded

So this could be an immensely valuable experience, it sometimes sounded like he was trying to convince me of why this is a good opportunity

Prep For Lunch

Prep For Internship

The real goal is:

Build the ability to independently design, test, and validate trading signals.

That requires **three capabilities**:

1. Mathematical intuition
2. Statistical modeling ability
3. Research engineering (coding pipelines)

Below is a **very concrete learning system** that tells you **exactly how to learn each skill**, what resources to use, and what outputs you must produce.

The Learning Philosophy

You will learn using **the Research Loop**:

Step 1
Learn concept

Step 2
Rebuild example from scratch

Step 3
Modify it

Step 4
Test it on financial data

Step 5
Write a short research memo explaining results

This loop forces understanding.

Core Resources (Do Not Add More)

You only need five.

Statistics

- *Statistical Inference – Casella & Berger* (selected chapters)

Time Series

- *Introductory Time Series with R – Cowpertwait*

Finance

- *Expected Returns – Antti Ilmanen*

Practical Quant

- *Hands-On AI Trading – Ernest Chan*

AI Applications

- *Generative AI for Trading – Ernest Chan*

These cover everything needed.

Daily Learning Structure (3 hours)

Every study session follows the same structure.

Block 1 (Concept Learning)

45 minutes

Read textbook section.

Goal
Understand theory.

Write a **one-page explanation in your own words**.

Block 2 (Rebuild Example)

60 minutes

Take an example from the book.

Rebuild the code **without copying**.

Example tasks

- regression
- signal generation
- factor construction

Block 3 (Modification Experiment)

45 minutes

Change something.

Examples

Change:

- lookback window
- portfolio weighting
- factor definition

Then observe results.

Block 4 (Research Note)

30 minutes

Write a short memo.

Structure:

Hypothesis
Method
Results
Interpretation

This builds real research thinking.

Phase 1

Statistics Foundation (Weeks 1–3)

Focus on **understanding statistical inference deeply**.

Topics

Probability distributions
Expectation and variance
Sampling distributions
Hypothesis testing
Confidence intervals

How to learn:

1. Read concept from Casella & Berger

2. Simulate it in Python

Example

Simulate coin tosses to understand sampling distribution.

Example code idea

```
import numpy as np
samples = np.random.binomial(1, 0.5, 1000)
```

Then analyze mean and variance.

Output

Short memo:

“Why sampling variability matters for trading signals.”

Phase 2

Regression and Factor Modeling (Weeks 4–6)

Topics

Linear regression
Multivariate regression
Factor models

How to learn

Take a factor hypothesis.

Example

Momentum predicts next-month returns.

Steps

Load stock returns

```
returns = prices.pct_change()
```

Compute momentum

```
momentum = prices / prices.shift(252) - 1
```

Run regression

```
import statsmodels.api as sm
model = sm.OLS(next_returns, momentum)
```

Interpret

- coefficient

- t-statistic
- p-value

Output

Research memo:

“Does momentum predict future returns?”

Phase 3

Time Series Modeling (Weeks 7–9)

Topics

Autocorrelation
AR models
Mean reversion
Volatility clustering

How to learn

Use real financial time series.

Example experiment

Test whether SPY returns are autocorrelated.

Steps

Compute lag correlation.

returns.autocorr(lag=1)

Interpret result.

Output

Research memo:

“Evidence of autocorrelation in equity returns.”

Phase 4

Strategy Construction (Weeks 10–12)

Now use **Ernest Chan books heavily**.

Each chapter becomes a **research experiment**.

Example

Mean reversion strategy.

Steps

1 Define signal

z_score = (price - rolling_mean) / rolling_std

2 Generate trades

3 Simulate portfolio

4 Analyze performance

Metrics

Sharpe ratio
Drawdown
Turnover

Output

Research report.

Phase 5

Regime Modeling (Weeks 13–16)

Now you start building **the kind of models relevant to the fund idea**.

Topics

Volatility regimes
Hidden Markov Models
Regime switching models

How to learn

Implement simple classifier.

Example

Volatility regime.

Compute rolling volatility.

vol = returns.rolling(30).std()

Classify

High volatility if above threshold.

Then test:

Which strategies perform better in each regime.

Output

Research report:

“Strategy performance conditional on volatility regimes.”

How You Should Use the Chan Books

You should treat them like **labs**.

Each chapter

1 Rebuild example code

2 Modify parameters

3 Test on different assets

4 Write research memo

Do not just read.

You must **experiment**.

Output You Must Produce

After this plan you should have:

5–10 research memos

Example titles

Momentum Predictability in US Equities
Mean Reversion in Large Cap Stocks
Factor Interaction Between Value and Momentum
Volatility Regimes and Strategy Performance
Dynamic Factor Allocation

These become your **quant research portfolio**.

What Makes You Stand Out

Not coding.

Not AI.

Your value will come from being able to say:

“Here is the hypothesis, here is the statistical evidence, and here is the portfolio implication.”

That is what PMs actually care about.

Proposal

How to Explain Your Work

You want to frame it as **three research streams** that support the new fund idea.

This shows:

• structure
• independence
• alignment with the PM's thesis

Version You Could Tell Them

"I've been organizing my work into three main research tracks so that everything I build connects back to the fund's strategy."

---

1. Factor and Signal Research

"First, I'm studying the behavior of different equity factors like value, growth, and momentum to see how predictive they are across different time periods.

The goal here is to identify signals that have statistical evidence behind them and could contribute to alpha generation."

What you are actually doing

• regression testing
• factor return analysis
• correlation analysis

How to explain it simply

"I'm testing which factors actually predict returns and under what conditions they work best."

---

2. Market Regime Detection

“Second, I’m exploring ways to classify the market environment so we can dynamically allocate between strategies.

For example, certain environments favor growth-oriented assets while others favor defensive or value-oriented exposures.”

Methods you could mention

• volatility regime detection
• Hidden Markov Models
• trend vs mean reversion classification

Simple explanation

“I’m trying to identify signals that tell us what kind of market environment we’re currently in.”

---

3. Portfolio Construction Framework

“Third, I’m working on translating those signals into a portfolio framework.

Once we understand which signals work and what regime we’re in, the next step is determining how to allocate capital across different asset buckets.”

Focus areas

• factor weighting
• diversification
• risk control
• correlation management

Simple explanation

“I’m designing a structure that can translate signals into portfolio allocations.”

How You Connect Everything

Then you finish with something like this:

“The long-term goal is to build a research pipeline where we can generate signals, identify the current market regime, and dynamically allocate across strategies in a systematic way.”

This makes it clear you are working toward:

a systematic investment engine.

If They Ask What You Are Learning

You can say:

“I’ve been focusing heavily on statistical modeling and time-series analysis so that any signals we develop are properly validated before they’re used in a portfolio.”

This signals seriousness.

What This Communicates

Without saying it directly, this tells them you are:

• thinking like a quant researcher
• structuring work professionally
• focused on signal validity
• aligned with the fund's strategy

That is exactly how good interns differentiate themselves.

One Optional Addition (Very Strong)

You could also mention this:

“I’m documenting each experiment as a research memo so we can keep a record of what signals worked, what didn’t, and why.”

PMs love this.

It shows **scientific discipline**.

Call Notes

Call Flow (30–40 min)

Opening (3–5 min)

Let Aiden share what he and Patrick have been thinking. Then:

> "I've been thinking about how to make this concrete. Let me walk you through how I'd scope the first 16 weeks."

Middle (20–25 min)

Walk through in this order:

1. **Strategy Universe** — Start narrow: momentum in growth stocks, specifically semiconductors + the supply chain. Mention the second-order plays (packaging, power, cooling, uranium) to show depth.
2. **Regime Detection** — Three signals: HMM for probabilistic state, Hurst for persistence, ADX for trend efficiency. Explain the displacement/distance physics concept. Offer to show your ADX chart image.
3. **Simulation** — Monte Carlo isn't just validation — it's forward-looking. Regime forecasting using HMM transition probabilities + scenario overlays.
4. **Options Hypothesis** — This is the differentiator. 95/5 split, convexity on high-conviction. Testable hypothesis with clear break-even analysis.
5. **Proof of Execution** — "I've already built a trend detector for gold using EMA curvature. Same signal logic applies. I can show you if useful."

Logistics (5–10 min)

Wait for him to raise it, or near the end:

> "I want to make sure we're aligned on logistics — hours, timeline, comp. What were you thinking?"

Key Phrases to Use

Moment	Exact Phrasing
Opening	"What's been on your mind since we last talked?"
Narrowing	"I'd start with one strategy class, one sector — then expand once we validate the pipeline."

Regime	"The bottleneck has moved from chips to infrastructure. That's where the second-order alpha is."
Simulation	"Monte Carlo isn't just for validation — it's for estimating future regime probabilities."
Options	"The hypothesis is that 5% in LEAPs on max-conviction names adds convexity without material drawdown risk."
Proof	"I've already built a system like this for gold. I can adapt it."
Hours	"What were you thinking in terms of weekly commitment?"
Comp	"I was thinking $35–40. Open to a lower base with discretionary bonus tied to whether the research gets deployed."

Structure

Quantitative fund research framework

Agilith Capital · 16-week systematic alpha research pipeline

Core hypothesis

Different strategies perform best under different market conditions. A systematic framework can identify these relationships, detect current regime, estimate future regime probabilities, and dynamically allocate capital as conditions evolve.

Phase 1: Strategy universe

Momentum in growth stocks with structural tailwinds

PRIMARY
Semiconductors
Direct AI infrastructure plays

SECONDARY
AI supply chain
Second-order bottleneck plays

Layer	Thesis	Tickers
Semis	Under-the-radar AI plays	CRDO SIMO SYNA
Packaging	CoWoS sold out through 2027	AMKR ASX
Power	5-year grid interconnect queues	VRT ETN
Cooling	Liquid-to-chip now mandatory	VRT MOD
Nuclear	Hyperscalers signing PPAs	CCJ CEG
Copper	500k+ tons/year by 2030	FCX

Phase 2: Regime detection

Multi-signal fusion for regime classification and forecasting

Hidden Markov Model
Probabilistic state inference from returns + volatility. Outputs regime probabilities + transition matrix.

Hurst exponent
Measures time series memory. H > 0.5 = trending, H < 0.5 = mean-reverting. Physics-based signal.

ADX
Displacement / distance ratio. Measures trend efficiency. Real-time strength indicator.

↓

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Regime → Strategy mapping

- Trending → Momentum
- Mean-reverting → Pairs
- High-vol → Reduce exposure

Phase 3: Forward simulation

Monte Carlo validation + scenario modeling + regime forecasting

Monte Carlo engine

- Block bootstrap (preserve autocorrelation)
- Regime-conditional path simulation
- Jump diffusion for tail events
- Robustness distributions + confidence intervals

Scenario + regime forecast

- Event impact models (Fed, earnings, geopolitical)
- HMM transition probabilities forward
- Scenario-adjusted regime forecasts
- Sentiment integration (optional LLM layer)

Phase 4: Portfolio construction

Dynamic allocation with convexity overlay

70%	25%	5%
Core equity	Satellite	LEAPs

Options overlay hypothesis

Can a small allocation to LEAP calls on high-conviction positions materially improve risk-adjusted returns through convexity? Testing: 12–18 month expiry, 10–20% OTM, regime-conditional deployment.

Technology stack

Data	Signals	ML / Regime	Backtest
`yfinance pandas sqlalchemy`	`ta-lib numpy scipy`	`hmmlearn sklearn`	`vectorbt quantstats`

Simulation	Options	Viz	Testing
`ray joblib`	`py_vollib mibian`	`plotly streamlit`	`pytest git`

16-week execution roadmap

Weeks 1–4 — **Foundation**
Data pipeline · Signal construction · Backtesting engine · 3 baseline strategies validated

Weeks 5–8 — **Regime detection**
HMM implementation · Hurst + ADX integration · Signal fusion · Strategy-regime mapping

Weeks 9–12 — **Forward simulation**
Monte Carlo engine · Scenario modeling · Regime forecasting · Full simulation pipeline

Weeks 13–16 — **Integration + allocation**
Dynamic allocator · PCA/factors · Options overlay · System testing · Documentation

Proof of execution

EMA curvature trend detector (built for gold)

First derivative of EMA → trend direction. Second derivative → trend acceleration. When both positive and increasing → high-probability momentum entry. Same signal logic transfers to equity momentum strategies.

Working system | Can demo on call

Parallel track: Portfolio assistant LLM

Supporting Aiden on knowledge graph + RAG system for research workflow

Schema for investment theses	RAG retrieval prototype	Knowledge graph architecture

Many different paths, lot of time and resources, not hitting everything,

Which are the most important ones

What are the overfitting strategies does it come again, market regime comes again

Two main priorities -

Apply current quant models & fundamentals -> US equities
Apply predicting market regimes

What resources -> Cloud compute resources, data packages, what do we want, need resources, how to predict these things, look at graphs, numbers, and other things

Define resources that we need, think of the type of data, are there available dtaa packages, social media, hard numbers

Data inputs and compute

Prediction markets -> chain of events, end result of chain of events, fundamental piece of data, and act before almost everyone

April 2nd Meeting

Quantitative Modeling -> very important

They want me to come as well

He has been going to many conferences to get more context

Quant modeling and then the LLM analyst

Subsections

Advantage -> improved coding capabilities, reasoning capabilities, small firm, don't need to market make, small cap -> more caution, but not moving markets

Signifcant pofit even if late

Me -> quant modeling

2 phases

Basics repairing gaps of LSEG -> data from them, fixing the quantitative modeling, take their ideas, root data, and modifying the models, order in terms of their thinking

Take data and then find trends

Success metric ->

Phase 2 -> signaling for in-depth research, we have quant models, now know, when is it not enough, when do we need to tell analyst smth is happening

Which combination -> movement is starting to happen, key for AI analyst side, take impetus and run deeper analysis, create a report into source document, sentiment

It is like a screenshot in time, over time, key moments of stock and industry

Builds mapping of market, when are these key moments in time? Important company events

Goal -> not doing everything, more -> 50-50, interested and useful to them

Step 3 -> portfolio balancing -> fairly solved problem, just implement the correct parts for it, drawbacks exit in llms

Money flow analysis -> insight into what players are trading a stock, to predict what will happen, not all action will be reflected in the price movement, order flow -> used to work, not sure now,

wanna check over here for alpha, big player -> key information, more analysis, search for extra alpha -> qualitative plan

SQL database and APIs -> creat the models as well go, how much can we feed into LLMs

LSEG -> provided models, some are broken, mostly made for american market, specific models that don't exist or are bad quality, some what recreate

Primary goal is canadian equities, then US, then EU later

My part can be flexible, look at US equities as well

Where is the value, answering that question

When are we going into market wide shift, or a company wide shift, what did the CEO just change, what is the messaging from the companies -> value vs growth vs mean reversion

Prediction markets for sentiment

Models don't work when regimes shift

Goal is we want the quant model short term model
LLm -> mid term
Long term

One more component -> looks for stock patterns, indicate where stock patterns

GenAlpha, AI alpha -> LLM, similar to our AI alpha, find alpha for quantitative models, strtgeies are fairly standardized

Cost vs predictive power increase

Pretty big fund -> very beneficial, worth the extra money

Resources Cost-Benefit Analysis

My focus: Momentum in US growth Stocks, using quantitative and fundamental techniques to identify good ways to make alpha

Even if the stock patterns have any alpha

News Sources

Data

Compute

2nd thing is regime detection, hwo amm I going to go about classifying regimes accurately, how to incorporate sentimenta analysis into it

AI tools

Starting point

Focus on estimate of cost and justified increase in how much alpha

Prediction Markets data

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The goal here is to identify signals that have statistical evidence behind them and could contribute to alpha generation.”

What you are actually doing

• regression testing
• factor return analysis
• correlation analysis

How to explain it simply

“I’m testing which factors actually predict returns and under what conditions they work best.

This is focusing on US equities with structural tailwinds

---

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ADX
Displacement / distance ratio. Measures trend efficiency. Real-time strength indicator.

And additionally, I think if if before the 10th you'd be able to send us something, even if it's preliminary about your like same thing as you sent the one pager here, send something similar of like your data that you want to look into the resources.
You need all of that stuff and an overall cost benefit structure, if not a theory.

I want, I want this much compute power. I want this this model or these series of models to help me or like and then looking. I'll send you that Alpha Gen. paper as well as well and their product and you can see if that's something that that you think would

But kind of just if something is very expensive, have a justification of why it's useful of of like at least a theory of why this is needed and and if it's if it's fairly cheap, cheap being, I'd

say anything under.
1000 or anything under $5000 is cheap that then you can kind of just say this is an experimental thing. This is like I want to experiment with this and this is my theory behind it. So they can be a loose cost benefit structure but.
Having something there that shows you've thought about this and as well As for us to then take in and do some of our own researches as to whether we agree with you or not on each of them, I think that would be super beneficial.

And and that that that can be that can be fairly rough like like I obviously you won't know the you won't know all of these things until you you actually test them but but just what your thoughts are on on each of the them they can be quite rough.

Data set or resource we can use in the meantime to start to get that proof of concept and then like, all right, once this is proving something for us, then we can invest in a more expensive one. But don't let cost be your main factor here in terms of deciding what is. Let your ideas kind of.
Ruminate as what data can help provide these these categories of of what I'm looking into.

The amount of of performance and predictive power to to improve stock performance by something like like an extra 5% of 5% on a year round basis.
4.
We would be able to justify quite a bit of money like in the in the probably like $50,000 range type thing.
I, I, I'm, I'm kind of speaking out of my *** years, but but this is based off the quick math I'm doing in my head. I don't know what their actual, um, uh, how much they're actually willing to to spend on any given thing and it's gonna have to go through like we're gonna have to propose it.
Talk about the specifics there. There's gonna have to be serious analysis of what is our predictive performance increase from including this versus what is our what is the cost here, but you're playing with with a a potentially a decently sized fund.
That that if it is really beneficial to have this as an extra marker, it's worth the extra money. So I think you we it could be anything from like 1000 to.
To 10s of thousands, let's say depending on how value, how valuable we can prove it will be because that is the other thing of like like this is a little bit of experimental as well. So maybe what we can do as well is maybe we can't, we don't know that it'll be, it'll be valuable off the start, but maybe there's a less expensive data.

And I'm not making any promises about like what will actually get approved, but it's it's worth it for you to have like say with this data, I think I'll be able to achieve this and with

this data I think I'll be able to achieve this or with this resource I'll be able to achieve this. If you're able to provide us with that, then we can make a decision.

But but there is flexibility in terms of capital to assign it. If you think this one component is going to be super valuable, then then we can look into that of of what will be the the cost risk scenario here.

Exactly, exactly. And as we're looking to define alpha, it's important for us to kind of have a theory as to where this chain of events like like like what is the end result of a chain of events and how can you go from really like your most fundamental piece of data to your end?
And alpha prediction essentially and be able to act on it before the majority of the market gets there. It's OK if it's if you're not the first first person, but you basically you want to you want to be there for as long as that trend happens essentially. So this is a.
This is AI think an important component for for you to to take some time and think about and I think in our next meeting and I don't want to hold you up forever here, but I think in our next meeting what?
Would be super valuable is to is or like even before our next meeting for you to take your one or two prime ideas and feel free to change them after this call. It's OK if if it doesn't stick to what you said in this call as like these new ideas entered and figure out.
What you need to actually accomplish them. You have a good strategy behind them. You have a good idea. What data do you want? What compute resources do you want? And I'll throw a couple of ideas at you as well. I'll put together a list of things after this call for that you could consider the.
Cal sheet and stuff like that is is one that could be interesting. Another resource that could be interesting is there was a research paper put out November or something like that and you can feel free to look into this as well. It's called.
What's it called? Gen. Alpha or something like that? Or L AI Alpha? No, something like that. I'll I'll I'll I'll I'll forward it to you. It's essentially it's a.
It's an LM that's going to be designed pretty similar to how we're probably picturing our architecture, but it's specifically used to help assist a analyst find alpha for quantitative models.
That it basically will assist in the in all of these back tests because they're fairly standardized as you're saying like these strategies are fairly standardized. So when you're trying to find alpha, this essentially will help you bounce ideas off as well as do the testing very quickly. So that could be a resource that you're like, hey, if we can afford this.
Then this would be useful. So I'll forward stuff like that to you. And if you essentially just do your best of don't be try your best to not be ridiculous about we're not a billion dollar fund here.

Especially if if it's like a third order result like you were saying where we get where like there's these major stocks that are appearing a lot of growth, but less like you can to give a great example is straight up Hormuz is closed that affects helium.
What's the likelihood of the straight of hormose opening in terms of its effect of helium, which it then turns into silicon chips, right? Like.

Like if you're looking for additional alpha and you're you're looking for how to predict these things as well as some of the best people on the market, um, it will probably be important to not just be looking at these graphs themselves, these hard numbers, but potentially be looking at other data sources, whether that be.
News, whether that be sentiment like market sentiment analysis, whether that be whatever you're there's there's a million alternative data sources as I've so discovered over the last month or so. There's millions out there and I can kind of.
I can, I can, I can refer you to some that I've seen, but a quick Google search will bring out a million of alternative data sources. I think what would be beneficial or what will be beneficial, what we kind of want from you for the next time that we meet is.
If we can start to define these resources that you need because we like like like you got a great plan here, you got some good ideas of of what to apply to the market and like we're we're very interested and I think where we go from here is.
Let's let's see what we can actually grab and don't worry too much about about the cost of what things are. Just try to come up with ideas of OK, this is the type of data that I need.
Are there available data packages for this stuff if you want to pull things from?
Social media or if you want to pull things from like just work with hard numbers that like whatever your strategy is, it's fine. But we will need to start to define what we want to do in terms of data inputs and the compute required to run these things as we start to design this plan of of as you look in.
These offers.

Um, so for the first one, um.
I mean it would probably be useful like for us to to just know in terms of these quantitative models, how will they work in general for the US market that like the quantitative models we're already using as well as as any theories that you have going forward that.
With that one question that I think is going to be useful for for like or that not not is going to be it might be useful and it will be useful. Something that we need to answer is what? Type of data or what data do you think you'll need for all of this as well as what resources do you want? We have, we have flexible capital on our side to invest in cloud compute resources or local compute resources. Probably we'll go cloud compute.

But like we have, we have resources to to invest. If you want specific data packages that you think would be valuable, these are things that that are going to be.

Circumstances applied. How well does this fit to all of those? Does it also work fit to them? Is this something that's just really good in this time period? OK, so if I can summarize your kind of two main priorities here, it would be applying.
Applying like our current quantitative models that we're using as well as some of this analysis to the US markets as well as predicting the regime that we're in, right. Those are the kind of the two main things. Fantastic.

Thought about a lot because I was like, yeah, everyone's like, oh, you have look ahead bias, you have overfitting. But I'm sure the strategy did something well in that time period for it to make that much money. And maybe if that condition appears again, then maybe I'll make money again, so.

Aiden Horan 53:46
Yeah, that that would that would have to be a theory that you test against like like you basically you have to come up with a theory of OK, what are the circumstances that applied to this time period that allowed it to overfit? OK, if let's look for every point in history where those where those same.

Yeah, I think the first thing which I'll do is very closely aligned with what we've already done, but in a different market. So I know you guys focus on Canadian equity. So if I take that momentum play idea and mostly it's mostly.
Value based right now, but if we take some of those value fundamentals and apply them to these high growth stocks in the US equity market, which have like big tailwinds right now because of all the A I developments and everything and everyone's looking at these stocks. So like I think there's there's like.
Is like there are two things to this. Like when every everyone's looking at a stock, either the move has already happened or when everyone's looking at a stock, the move hasn't happened. But like people are starting to get inclined to like moving it upwards. So I think finding the second subset.
Of of these stocks is like probably what I would focus on 1st because then I can also leverage what we what Patrick has already built over like 20 years of experience to apply his value based fundamentals to identify those high growth companies which have good fundamentals.
I think that would be the first thing which I would tackle and how to use our quant models to find find like anomalies of when those when those stocks are indicating to us that

probably right now is a good time to go and do more research or right now is a good time to buy or sell and then how can we?
Build a portfolio and then the second thing would probably be the regime detection because I think that's gonna fall into every single thing which we do in the future. If you can't, if you can't like classify what market condition you're in right now, no matter how many strategies you have.

Yeah, yeah, yeah, absolutely. And and it's it's it's I think I think looking into leveraging or extra leverage on on very strong like on on very strongly held opinions or strongly held views.
Where a potential stock or market is going to go, I think it makes complete sense. I think it makes sense. I and I think that's something that that Patrick also does like like putting a little bit of extra leverage. I don't think he does too much of.
Of like being super over leveraged, he's he's not a super like a a crazy risk guy, but he does. He certainly plays with with putting extra leverage on these strongly held positions and there's a few.
There's a few components that we can kind of utilise like there's each of these quant models have we can figure out how much they kind of align with each other versus they can be used as independent signalling. But there's a few different things that we can use as signalling for the future performance of stock, another one that we have access to.
Is things like insider trading so we we can get reports of of all of the insider trading that is happening and if you Patrick was was explaining that one of his his go tos is if you see like a.
Um, if you see basically investments of buying a stock in the C-suite, it's a very good product like predictor that the stock is gonna be going up. Um, that almost certainly if someone if the people in the C-suite are buying it, then they know something that.
That you don't know or they because they're gonna be the ones that are the experts on their company. At the end of the day, even if even if they're just using public information, they probably know something you don't because they are the expert on their on their feet, like in their industry, they are the expert in their company.
So there it's a very good indicator of something like that is happening as well as we have the quantitative models as well as we can use these these graph analysis essentially. And basically what Patrick was saying that he does is he looks for a triangulation of these three things to kind of come to a conclusion. But if there's a lot.
Factors that are all pointing in the same direction. This looks like it could be a very profitable thing. Leverage it. We might like it might. It makes sense to play with that type of stuff. Now what I want to kind of this is also fantastic by the way.

I attached the transcript for our call to this email. Additionally, here are some of the resources we discussed as potential places we can find alpha.

Alpha gpt as an alpha creation assistant: github: https://github.com/parthmodi152/alpha-gpt, product: https://alphagpt.vercel.app/, white paper: https://arxiv.org/abs/2308.00016

Use of prediction markets to explore 2nd 3rd or 4th order effects

Some other things ive seen that look interesting are here as well: increase alpha does language processing of sec filings: https://increasealpha.com/, news sentiment analysis (I know there are a number of products out there for this), 3AI builds models on top of the same LSEG data that we are using: https://www.3ai.co/

In terms of the data and resources you are looking at obviously the cheaper the resource is, the better and easier it will be to approve but if you believe a certain resource can be the difference between 10% returns and 15% returns then it could be worth quite a bit to experiment with it. Let me know if you have any questions or you want to run any ideas by us to gauge whether they are feasible.

Potential Alpha

1) Infrastructure Constraint Nowcast

Like you are 5

AI companies want to build giant data centers.

But they cannot always do it fast, because they need:

- enough electricity
- transformers
- switchgear
- cooling
- grid connection

So this idea says:

when the world is short on the "stuff needed to build AI," the companies that already own or sell that stuff may win first.

Not the flashy AI app companies.
The shovel sellers.

Why it might work

Most people hear "AI demand is strong" and buy the obvious names.

But this idea says the better question is:
who benefits when AI demand runs into physical traffic jams?

That can shift earnings power toward companies with scarce equipment, long backlogs, pricing power, or reserved capacity. The blueprint explicitly frames this as turning AI from a demand story into a supply story.

What you need to test

You need to test whether rising "constraint severity" predicts future stock returns.

Test it in steps:

Step 1: Build a constraint score
Use company text and public data to score how tight the bottlenecks are.

Inputs:

- earnings call transcripts
- 10-K / 10-Q filings
- keywords like transformer, lead time, interconnection, power availability, liquid cooling, curtailment
- public grid / transformer shortage information

Step 2: Link each stock to the bottleneck
For each stock, estimate:

- does it sell scarce equipment?
- does it depend on external power connection?
- is it a beneficiary or victim?

Step 3: Backtest
Each week or month:

- long highest-quality beneficiaries
- short names whose valuation assumes smooth AI buildout but whose language shows dependency on delayed infrastructure

Step 4: Validate
Check:

- hit rate
- spread return
- turnover
- factor exposure
- whether alpha survives after controlling for size, sector, momentum, and valuation

What you need to execute it

Minimum stack:

- universe of relevant industrials / utilities / cooling / electrical / data center suppliers
- filings + transcripts
- public infrastructure references
- simple scoring engine
- monthly rebalance system
- risk model so you do not just accidentally buy “industrials beta”

Free / cheap / expensive options

Free

- SEC EDGAR for filings
- company IR pages for some transcripts / decks
- DOE / CISA / public grid reports
- Python + pandas + SQLite / Parquet
- manual keyword dictionaries

Cheap

- cheap transcript provider or scraped transcript archive
- OpenAI / Claude API for text classification
- Runpod for burst embeddings
- small feature store on local machine or cheap cloud storage

Expensive

- premium transcript feed
- expert network / infra industry channel checks
- alternative data on supply chain or lead times
- full research pipeline with orchestration and monitoring

My view

This is one of the best ideas in the document.

Why:

- non-obvious
- medium horizon
- economically real
- not purely narrative
- scalable into a repeatable system

Main danger:
you accidentally build a cool thematic story instead of a true predictive signal.

---

2) Liquid Cooling Inflection Capture

Like you are 5

At first, AI computers can be cooled by air.

But after they get too powerful, air is not enough.
Then companies must switch to liquid cooling.

This idea says:

find the exact moment when customers go from “maybe liquid cooling” to “we now need it.”

Then buy the winners before revenue fully shows up.

Why it might work

The key is not “liquid cooling is important.”
Everybody knows that.

The key is:
when does the customer cross the threshold where liquid cooling becomes required, not optional?

That transition is nonlinear. Once rack density gets too high, spending can shift fast. The blueprint says the edge is detecting that inflection quarter from customer language and backlog / margin signals before revenue recognition fully reflects it.

What you need to test

You need to detect transition timing.

Step 1: Build a state label
For each company-quarter, classify:

- air adequate
- hybrid
- liquid required

You can do this from transcript language.

Step 2: Add confirmation
Look for:

- backlog growth
- segment margin improvement
- capex mix shifting toward high-density solutions
- customer references to power density

Step 3: Build long-short baskets
Long:

- beneficiaries of liquid cooling adoption

Short:

- firms exposed to air-cooling ceiling risk or overhyped names where transition is not actually confirmed

Step 4: Backtest around the state change
Best test:

- when company moves from state A to B
- what happens over next 20, 40, 60 trading days?

What you need to execute it

- technical language dictionary around cooling / rack density
- transcript parser
- some method to classify each quarter
- structured segment data
- liquid tradable universe
- regime filter so you do not force trades in bad tape

Free / cheap / expensive options

Free

- public company presentations
- earnings call text if accessible
- manual labeling in a spreadsheet for small universe
- simple rules like “mentions of liquid cooling + density + retrofit + deployment”

Cheap

- LLM classifier for quarter labels
- basic transcript feed
- simple dashboard for state transitions
- GPU bursts for embeddings if needed

Expensive

- engineering expert input
- paid infra / data center industry research
- broader transcript coverage
- event-driven trading around hyperscaler capex dates

My view

Very interesting, but more fragile than idea 1.

Why:

- timing matters a lot
- crowded obvious names may already price it
- technical shifts can change fast

Still, this is a strong submodule if you keep it narrow and specific.

---

3) Transformer and Interconnection Delay Transmission

Like you are 5

A company says:
"We are building a huge AI data center."

But that does not mean it turns on tomorrow.

It still needs:

- transformer delivery
- grid approval
- interconnection
- actual power energization

This idea says:

some companies are priced like the lights will turn on soon, but real-world delays may push that out by quarters.

So short the over-assumed winners, long the bottleneck suppliers.

Why it might work

This is a timing mismatch trade.

The market may price the story:
"AI capex is big."

But realized earnings depend on when projects are actually energized. The blueprint highlights that delay information often sits outside normal financial datasets, such as queue reports and grid bottlenecks.

What you need to test

Step 1: Build a delay-risk score
Use:

- transcript mentions of interconnection / energization / power timing
- regional grid queue information
- capex guidance that depends on power availability

Step 2: Split names into two groups
Group A:
stocks whose valuations assume near-term capacity ramp

Group B:
stocks with pricing power because everyone else is delayed

Step 3: Pair-trade or basket-trade

- short “too early” names
- long bottleneck winners

Step 4: Test timing windows
Very important:

- 1 month
- 3 months
- 6 months
  Because delays may be slow-moving.

What you need to execute it

- regional mapping of data center exposure
- power / queue / interconnection public data
- transcript text tagging
- valuation framework
- pair-trade risk controls

Free / cheap / expensive options

Free

- public queue reports
- utility / grid operator documents
- DOE / public reports
- company filings and calls
- manual regional mapping

Cheap

- LLM extraction of “power contingent” language
- simple internal database of projects and expected energization references

- cloud notebook for monthly refresh

Expensive

- paid infrastructure intelligence
- commercial supply chain data
- expert calls with electrical contractors / utility consultants

My view

This is excellent if you care about **timing alpha** rather than pure thematic exposure.

It is hard, but good hard.

Big weakness:
the data can be messy and laggy.
You need discipline or it becomes macro fan fiction.

---

4) Margin Before Revenue Recognition

Like you are 5

A company uses AI inside its business.

At first, it may not sell more stuff.

But it may become cheaper and faster to run:

- support
- coding
- sales enablement
- customer service
- internal workflows

So profits improve before revenue does.

This idea says:

buy companies where AI is already cutting costs in a real way, even if Wall Street is still waiting to see revenue growth.

Why it might work

Many investors look for AI in top-line growth.

But enterprise AI often shows up first in:

- lower costs
- higher operating leverage
- better productivity
- higher revenue per employee

The blueprint's point is that screens may misread this as maturity or cyclical leverage, instead of structural AI productivity.

What you need to test

Step 1: Build structured productivity features
For each company:

- SG&A / revenue trend
- R&D / revenue trend
- revenue per employee
- operating margin acceleration
- maybe headcount efficiency

Step 2: Add text evidence
Look for language moving from:

- pilot
  to
- scaled
  and look for specifics:
- named deployment
- measured time saved
- actual workflow change

Step 3: Exclude fake cases
Very important:
separate real AI productivity from:

- layoffs
- temporary cost cuts
- one-time restructuring
- accounting distortions

Step 4: Backtest post-earnings
This likely works best around earnings and then over 8 to 16 weeks, which matches the blueprint.

What you need to execute it

- financial statements
- employee count data
- filings + earnings calls
- event-study framework
- clean accounting adjustments

Free / cheap / expensive options

Free

- SEC filings
- company annual reports
- manual extraction for a small universe
- simple spreadsheet or Python factor model

Cheap

- LLM to detect “pilot vs scaled” language
- transcript source
- basic feature pipeline

Expensive

- high-quality fundamentals feed with segment detail
- better workforce / org-change data
- more advanced text pipelines

My view

Very strong.

Why:

- grounded in real economics
- feasible at Tier 1
- not dependent on exotic data
- closer to actual earnings

This is one of the first ideas I would seriously prototype.

---

5) AI Engagement Surprise and Mispricing

Like you are 5

Some companies suddenly start talking a lot more about AI.

But not all AI talk matters.

This idea says:

do not look at who talks most about AI. Look at who changed their AI talk a lot relative to their own past and their peers, and then confirm it with real actions.

So the signal is “surprise,” not raw buzz.

Why it might work

Raw AI mention counts are now crowded and dumb.

The smarter signal is:

- how unusual is this change for this company?
- how unusual is it versus similar firms?
- is there real follow-through?

The blueprint recommends validating with capex, R&D changes, margin-before-revenue effects, or bookings proxies.

What you need to test

Step 1: Build AI text metrics
For filings:

- AI term counts
- weighted frequency
- embeddings / topic similarity

Step 2: Create surprise score
Compare current quarter to:

- company’s own history
- peer group history

Step 3: Add confirmation filter
Only keep names where at least one real business signal confirms the text:

- higher capex
- R&D reallocation
- actual product launch
- margin shift
- bookings proxy

Step 4: Long-short backtest
Long:

- real high-surprise names with confirmation

Short:

- AI mention inflation with no operational proof

What you need to execute it

- filing parser
- peer group construction method
- text scoring pipeline
- confirmation feature layer
- event / medium-horizon backtest

Free / cheap / expensive options

Free

- SEC API
- basic keyword counts
- manual peer groups
- cosine similarity from open-source embeddings

Cheap

- LLM or embeddings API for topic detection
- Hoberg-Phillips style text-based peer methods if replicated simply
- small cloud job for corpus refresh

Expensive

- enterprise NLP pipeline
- full-universe transcript + filing corpus
- richer alternative validation feeds

My view

This is good, but only if you are strict.

Raw “AI mention factor” is weak.
The whole edge is in:

- change, not level
- peer-normalization

- confirmation, not hype

If you skip those, it becomes nonsense fast.

My honest ranking of these first 5

For a serious summer research foundation, I would rank them:

Tier A

1. Infrastructure Constraint Nowcast
2. Margin Before Revenue Recognition
3. Transformer / Interconnection Delay

Tier B
4. AI Engagement Surprise
5. Liquid Cooling Inflection

Why:

- Tier A is more generalizable, economically grounded, and easier to test with your likely setup.
- Tier B is still useful, but either more timing-sensitive or more vulnerable to hype / crowding.

What I would build first if I were you

Not a full strategy.
A **research foundation stack**.

Build these reusable pieces first, because multiple ideas need them:

Module 1: Filing + transcript ingestion

- SEC filings
- earnings call text
- clean company-quarter database

Module 2: Text feature engine

- keyword counts
- finance-specific sentiment baseline

- “specific vs vague” classifier
- “pilot vs scaled” classifier
- constraint severity classifier

Module 3: Structured feature engine

- margins
- SG&A / revenue
- R&D / revenue
- capex
- backlog if available
- revenue per employee
- revision data if you have it

Module 4: Event-study harness

- earnings windows
- 20 / 40 / 60 day forward returns
- long-short spread testing
- sector-neutral / size-neutral checks

Module 5: False positive control

- walk-forward
- holdout period
- turnover and cost assumptions
- negative-result log

That matches the blueprint’s emphasis that the real edge at low budget is measurement, disciplined text + fundamentals fusion, and avoiding overfitting.

Next, I can do the next 5 ideas in the same format:

- Narrative vs Numbers Divergence
- Voice Stress Underreaction
- Analyst Revision Diffusion with Tone Gating
- Regime-Conditioned Sentiment and Attention
- CPI and FOMC Scenario Second-Order Trade

Good. These next 5 are more advanced and closer to “edge layering” rather than core signals. Some are strong. Some are easy to mess up.

I’ll keep the same structure and stay practical.

---

6) Narrative vs Numbers Divergence

("loud storytellers vs quiet executors")

Like you are 5

Some companies talk a lot.

Some companies just do the work.

This idea says:

buy the companies that quietly deliver results, and short the ones that talk big but don't deliver.

Why it might work

Markets reward good stories, especially in growth stocks.

But over time:

- results matter more than words
- hype gets punished if it doesn't convert

The blueprint calls this a "credibility spread" between narrative and delivery.

What you need to test

Step 1: Build a "narrative score"
From transcripts + filings:

- how much they talk about growth, AI, opportunity
- how specific they are (very important)
    - vague: "we are excited"
    - concrete: "we deployed X to Y customers and saved Z hours"

Step 2: Build a "delivery score"
From financials:

- revenue growth
- margin expansion
- bookings proxies
- guidance accuracy

Step 3: Compute divergence
Narrative acceleration − delivery acceleration

Step 4: Backtest

- long low narrative / high delivery (quiet executors)
- short high narrative / low delivery (story sellers)
- test around earnings cycles (4–10 weeks)

What you need to execute it

- text pipeline (concreteness detection)
- financial metrics
- alignment logic (matching narrative vs numbers over time)
- event-based backtest

Free / cheap / expensive

Free

- filings + transcripts
- simple keyword + phrase rules
- manual labeling for small universe

Cheap

- LLM to classify “vague vs specific”
- simple scoring model
- lightweight pipeline

Expensive

- full NLP pipeline with supervised training
- large transcript coverage
- advanced guidance tracking datasets

My view

Strong idea.

But:

- very easy to overfit
- very easy to build a “vibes model”

Only works if:
you **force structure and measurement** into narrative scoring.

---

7) Voice Stress Underreaction

Like you are 5

When people talk, their voice gives clues:

- nervous
- confident
- unsure

This idea says:

managers might sound stressed or uncertain even if the words look fine. That hidden signal might predict future performance.

Why it might work

People cannot fully fake their voice.

Research shows:

- tone, pauses, pitch variability can carry information
- analysts may not fully price it in

The blueprint explicitly notes this as an underused signal because audio is messy.

What you need to test

Step 1: Get audio

- earnings call recordings

Step 2: Extract features

- pitch variability
- speech rate
- pauses
- tone shifts

Step 3: Compare vs text
Find cases where:

- transcript sentiment = positive
- voice = stressed / uncertain

Step 4: Add trigger
Only trade when combined with:

- earnings surprise
- guidance change
- analyst revision shifts

Step 5: Backtest
Event-based:

- post-earnings drift
- 20–60 day horizon

What you need to execute it

- audio data
- signal processing tools
- alignment with transcript text
- event framework

Free / cheap / expensive

Free

- some public call recordings (limited)
- basic audio libraries (Python)

Cheap

- partial datasets
- simple acoustic feature extraction
- small-scale testing

Expensive

- full earnings call audio dataset
- robust signal processing pipeline
- large-scale compute

My view

This is **not a priority** for you right now.

Why:

- data friction is high
- signal is fragile
- execution complexity is high

This is a **Tier 2–3 niche edge**, not a foundation.

---

8) Analyst Revision Diffusion with Tone Gating

Like you are 5

Analysts update their forecasts slowly.

They don't all react at once.

This idea says:

when analysts slowly upgrade or downgrade a stock, and the company's language becomes more concrete, prices may keep moving in that direction.

Why it might work

There is a known effect:

- prices drift after earnings and revisions

But this improves it by:

- tracking how fast revisions are changing

- combining with management language

The blueprint frames this as combining two underreaction channels.

What you need to test

Step 1: Build revision features

- change in analyst estimates
- acceleration (1st derivative)
- acceleration of acceleration (2nd derivative)
- dispersion (do analysts agree?)

Step 2: Add tone gate
Only take trades when:

- management language becomes more specific / operational

Step 3: Build signal

- strong positive revisions + strong tone → long
- weak or negative revisions + weak tone → short

Step 4: Backtest

- 20–60 trading days (classic drift window)

What you need to execute it

- analyst estimate data (important)
- transcript text processing
- feature engineering
- drift-based backtest

Free / cheap / expensive

Free

- very limited analyst data (hard to get properly)

Cheap

- partial datasets
- approximations via guidance + price reaction

Expensive

- full estimates dataset (LSEG, Bloomberg, etc.)
- high-quality revision history

My view

This is **very strong if you have data**.

If you already have LSEG access, this becomes:

- one of your best signals
- very grounded
- historically validated

This is a **serious contender for top 2**.

---

9) Regime-Conditioned Sentiment and Attention

Like you are 5

Some strategies only work sometimes.

Example:

- sentiment matters more when markets are scared
- less when everything is calm

This idea says:

only use your signals when the environment is right.

Why it might work

Most quant signals are applied all the time.

But reality:

- markets behave differently in different regimes
- signal strength varies

The blueprint emphasizes that sentiment works better in certain macro states and attention conditions.

What you need to test

Step 1: Define regimes

- volatility (VIX)
- trend (market breadth)
- liquidity (spreads, turnover)
- macro windows (CPI, FOMC)

Step 2: Tag historical data
Each time period gets a regime label.

Step 3: Test your signals inside each regime
Example:

- does sentiment alpha work better in high volatility?

Step 4: Condition trades
Only execute signal when:

- regime = favorable

What you need to execute it

- regime classification system
- mapping of signal → performance by regime
- gating logic in portfolio

Free / cheap / expensive

Free

- price data
- simple volatility measures
- basic regime classification

Cheap

- slightly richer macro features
- simple ML regime models

Expensive

- full macro + liquidity datasets
- advanced regime modeling frameworks

My view

This is not a standalone alpha.

This is a **multiplier**.

It improves:

- every other signal you build

You should build this early, but not as your main strategy.

---

10) CPI / FOMC Second-Order Trades

Like you are 5

When big news comes out (like inflation or Fed decisions):

Everyone reacts immediately.

But not everything reacts at once.

This idea says:

trade the delayed effects, not the obvious first move.

Why it might work

Everyone trades:

- “rates up → tech down”

But fewer people trade:

- how this changes:
    - enterprise budgets
    - infrastructure spending
    - AI adoption timing
    - second-order sectors

The blueprint emphasizes attention and delayed repricing after macro events.

What you need to test

Step 1: Build event calendar

- CPI dates
- FOMC dates

Step 2: Define scenarios

- CPI surprise up / down
- volatility regime
- pre-event positioning

Step 3: Identify second-order baskets
Example:

- constraint beneficiaries
- enterprise software vs infra
- adoption-sensitive names

Step 4: Backtest

- enter after event
- hold 3–15 days

What you need to execute it

- event calendar
- macro surprise measurement
- sector / basket construction
- short-horizon backtesting

Free / cheap / expensive

Free

- CPI + FOMC schedules
- price data
- simple surprise proxies

Cheap

- macro data feeds
- better surprise modeling

Expensive

- high-frequency macro reaction data
- institutional-grade event systems

My view

Interesting, but:

- very noisy
- sensitive to macro regime shifts
- hard to make stable

Better as a **supporting overlay**, not core alpha.

Resources Report

1. Market Data & Screener (MVP Setup)

Purpose

Market data is required to support:

- momentum-based screening
- backtesting and signal evaluation
- volatility inputs for regime classification

A screener is used to generate a **point-in-time, survivorship-aware universe** for historical testing.

MVP Market Data Sources

Alpha Vantage

- API-based access to price data, including intraday (1-minute) time series
- Free tier available; paid tier (~$49.99/month) provides higher rate limits (~75 API requests/minute)
- Strong fit for AI-native workflows due to ease of integration
- Used for: structured price data pipelines

Yahoo Finance (via yfinance)

- Direct Python integration enables fast and flexible data access
- Used for rapid prototyping and quick iteration of backtests

---

Primary Data Layer

Financial Modeling Prep (FMP)

- Broad, integrated dataset including:
    - price data
    - fundamentals
    - analyst estimates and revisions
- Analyst revisions API is particularly valuable for building a **sell-side sentiment / revision agent**, enabling automation of tasks traditionally requiring manual analyst review
- Pricing:

    - Ultimate plan: ~$49/month
    - Enterprise pricing available upon request

Rationale:
FMP provides the best balance of coverage, flexibility, and cost for MVP development, allowing multiple signals (price, fundamentals, revisions) to be tested within a single API.

Screener (Point-in-Time Universe Construction)

Quant Investing Screener

- Provides point-in-time screening capabilities, helping reduce survivorship bias in historical backtests
- Enables construction of realistic historical universes rather than relying on current index constituents

Pricing:

- Unlimited plan: ~£36/month

Role in System:

- Generates candidate universe for backtesting
- Ensures realistic historical simulation environment

Scaling Considerations (Not Required for MVP)

More advanced data providers such as Polygon, Bloomberg, FactSet, and Finnhub offer higher-quality, real-time, or specialized datasets. These are not required for initial signal validation and would only be considered once consistent alpha has been demonstrated.

TLDR: MVP Data Stack

Component	Provider	Monthly Cost	Role in System	Alpha Enablement
Market Data (API)	Alpha Vantage	$0 – $49.99	Price data, intraday access, API pipeline	Enables rapid prototyping and agent integration

Rapid Prototyping	Yahoo Finance (yfinance)	$0	Quick iteration and backtesting	High iteration speed, low cost
Core Data Layer	FMP	~$49	Prices, fundamentals, analyst revisions	High impact – supports multiple alpha signals (especially revision-based agents)
Screener	Quant Investing	~£36 (~$45)	Point-in-time universe construction	High impact – reduces survivorship bias in backtests

Note: | Future Upgrade | Polygon / LSEG / Bloomberg | $$$ | High-quality real-time / institutional data | Only relevant after signal validation |

2. Fundamental Data (MVP Setup)

Purpose

Fundamental data is required to support:

- margin-based signals (core alpha driver)
- growth and quality filters
- validation of financial performance trends

MVP Data Sources

SEC EDGAR (data.sec.gov APIs)

- Official source of company filings (10-K, 10-Q)
- Accessible via public APIs, enabling programmatic ingestion through Python
- Provides point-in-time financial disclosures

System Integration:

- Filings are ingested via Python pipelines
- Routed through a custom MCP layer
- Fed directly into the **Fundamental Signal Agent** for structured analysis and signal extraction

Used for:

- source-of-truth financial validation
- extraction of deeper financial context beyond standardized metrics

---

Financial Modeling Prep (FMP)

- Same core data provider as defined in Section 1 (Market Data)
- Provides standardized financial statements via API, including:
    - income statement
    - balance sheet
    - cash flow

Used for:

- rapid retrieval of structured fundamentals
- integration into feature pipelines for backtesting

Pricing:

- Premium individual plan: ~$49/month
- Enterprise pricing available via quote / consultation

Rationale

FMP enables fast access to structured financial data, while SEC EDGAR ensures point-in-time accuracy and completeness. Together, they provide both speed and reliability without requiring institutional datasets at the MVP stage.

Scaling Considerations (Not Required for MVP)

Institutional providers such as Refinitiv, FactSet, and Capital IQ offer normalized datasets, deeper historical coverage, and metadata. These would only be considered once signal validity is confirmed and data quality becomes a constraint.

Fundamental Data Stack

Component	Provider	Monthly Cost	Role in System	Alpha Enablement
Structured Fundamentals	FMP	~$49	Financial statements, margins, core metrics	Enables margin-based and quality signals
Raw Filings (API + MCP)	SEC EDGAR	$0	Point-in-time filings ingested via Python → MCP → agent pipeline	Enables deeper financial analysis and prevents lookahead bias

3. Analyst Estimates & Revisions

Purpose

Analyst estimates and revisions are used to support:

- detection of changes in market expectations
- revision-based alpha signals
- construction of a **sell-side sentiment / expectations agent**

This layer is intended to capture **underreaction to analyst updates**, which is a well-documented source of alpha.

MVP Data Source

Financial Modeling Prep (FMP)

- Same core provider as defined in Sections 1 and 2
- Provides:
    - EPS estimates
    - revenue forecasts
    - analyst revisions over time

System Integration

- Estimates data is ingested via API
- Used to construct:
    - revision momentum features
    - estimate dispersion signals (later stage)
- Fed into a **Revision / Sentiment Agent** to:
    - track direction and magnitude of estimate changes
    - combine with price and narrative signals

Rationale

FMP was specifically selected as the primary data provider due to its inclusion of analyst estimates and revisions, which are not available in many comparable low-cost APIs.

This enables:

- early testing of revision-based signals
- development of expectation-driven models
- expansion into sentiment-aware agent systems

without requiring expensive institutional datasets.

Scaling Considerations (Not Required for MVP)

Institutional datasets such as Refinitiv I/B/E/S or Bloomberg provide:

- deeper coverage
- true point-in-time estimate histories
- more complete analyst tracking

These would only be considered once revision-based signals are validated and require higher data fidelity.

MVP Analyst Stack

Component	Provider	Monthly Cost	Role in System	Alpha Enablement

Analyst Estimates & Revisions	FMP	~$49	EPS/revenue forecasts, revision tracking	Enables revision-based alpha and sentiment-aware agents

4. Text Data

Purpose

Text data is required to support:

- **Infrastructure Constraint Nowcast** (supply bottlenecks, capacity language)
- **Narrative vs Numbers Divergence** (hype vs execution)
- **Analyst Revision + Tone Gating** (confirmation via management language)

This layer enables extraction of **qualitative signals** not captured in price or fundamentals.

MVP Data Sources

SEC Filings (EDGAR APIs)

- Official filings (10-K, 10-Q, 8-K)
- Accessible via public APIs and bulk endpoints
- Provides timestamped, point-in-time corporate disclosures

System Integration:

- Ingested via Python pipelines
- Processed through MCP layer
- Routed into:
    - Constraint Agent
    - Narrative / Sentiment Agent

Used for:

- detecting infrastructure constraints (e.g., capacity, supply issues)
- identifying changes in company language over time
- extracting structured + unstructured signals

Earnings Call Transcripts (via FMP)

- Retrieved programmatically via FMP API
- Provides direct access to management discussion and Q&A

System Integration:

- Pulled into pipeline alongside fundamentals and estimates
- Processed via LLMs for:
    - tone analysis
    - concreteness detection
    - narrative evolution

Used for:

- measuring management credibility
- validating analyst revisions through tone gating
- tracking narrative shifts across quarters

News (Lightweight / On-Demand)

- Aggregated via:
    - web access (Google News, RSS)
    - public X (Twitter) accounts that publish real-time headlines (e.g., FirstSquawk)

Used for:

- event detection
- supplementary sentiment signals
- real-time context around major developments

NLP / Processing Layer

LLMs (Claude / OpenAI APIs)

Used for:

- summarization

- classification (e.g., vague vs specific language)
- signal extraction from filings and transcripts

Rationale

Text data is critical for extracting non-obvious signals related to:

- supply constraints
- management behavior
- narrative shifts

Using SEC filings and FMP transcripts provides a **high-quality, low-cost, API-integrated foundation**, while LLM-based processing enables structured signal extraction without requiring expensive proprietary datasets.

Scaling Considerations (Not Required for MVP)

Premium providers such as Bloomberg News, AlphaSense, and FactSet transcripts offer:

- faster and more reliable access
- cleaner formatting
- broader coverage

These would only be considered once text-based signals are validated and require higher fidelity or scale.

Component	Provider	Monthly Cost	Role in System	Alpha Enablement
Filings (API + MCP)	SEC EDGAR	$0	Structured + unstructured disclosures	Enables constraint and narrative signals
Earnings Transcripts	FMP	Included (~$49 total)	Management tone and narrative	Enables narrative + tone gating signals
News (Lightweight)	Web + X (e.g., FirstSquawk)	$0	Event detection and context	Supplementary sentiment signals

NLP Processing	Claude / OpenAI APIs	Usage-based	Text parsing and signal extraction	Converts text → structured features

Macro Data

Purpose

Macro data is used to support:

- **Regime-Conditioned Signal Engine**
- identification of market environments (risk-on / risk-off)
- filtering when signals are most effective

MVP Data Source

FRED (Federal Reserve Economic Data)

- Free API providing access to key macroeconomic indicators
- Extensive coverage of:
    - inflation (CPI, PCE)
    - interest rates
    - employment
    - growth metrics

System Integration:

- Pulled via API into Python pipelines
- Used to construct:
    - volatility / macro regime indicators
    - event-based filters (e.g., CPI releases)

Rationale

FRED provides high-quality, official macroeconomic data at no cost and is sufficient for constructing regime classification features at the MVP stage.

Scaling Considerations (Not Required for MVP)

More advanced macro data providers (Bloomberg, Trading Economics, proprietary feeds) offer:

- faster updates
- global coverage
- forecast data

These would only be required if macro signals become a core driver of alpha.

Component	Provider	Monthly Cost	Role in System	Alpha Enablement
Macro Data	FRED	$0	Inflation, rates, employment	Enables regime classification and signal timing

6. AI, Compute & Tooling

Purpose

This layer supports:

- agent-based signal generation
- development and research workflows
- backtesting, simulation, and iteration speed

Core Development Stack

LLM Platform (Primary): Claude Max

- Used as the **central platform** for:
    - research and analysis

    - coding and debugging
    - agent development
    - workflow orchestration

Role in System:

- Acts as a **single unified interface** for:
    - building agents
    - processing text data
    - generating structured signals
    - accelerating development cycles

Rationale:

- Consolidating workflows into a single LLM platform improves speed and reduces fragmentation
- Prior experience shows strong productivity gains when fully leveraging Claude's capabilities (chat, code, agents, skills)
- Other tools (e.g., ChatGPT, Cursor) are intentionally excluded to maintain focus and maximize efficiency

---

Development Environment: VS Code + GitHub

- Standard development stack for:
    - Python pipelines
    - backtesting frameworks
    - version control

Role in System:

- Code development and execution
- Repository management and reproducibility

Charting & Real-Time Analysis: TradingView (Pro Plan)

- Pricing: ~$50/month

Used for:

- real-time price monitoring
- chart pattern analysis
- validating signals visually

- integrating with Claude via MCP

Rationale:

- Provides fast feedback loop between:
    - systematic signals
    - real-time market behavior
- Bridges quantitative system with discretionary insight

Open-Source Tooling (Supporting Layer)

In addition to the core stack, the system will leverage open-source tools and libraries to accelerate development and reduce cost:

- **Python (open-source)** serves as the primary programming language for:
    - data pipelines
    - backtesting
    - feature engineering
    - agent integration
- Core Python libraries include:
    - pandas, numpy (data processing)
    - requests (API integration)
    - standard ML/NLP tooling as needed
- Selected open-source repositories will be used as reference implementations and building blocks:
    - **alpha-gpt** (shared internally) for AI-driven financial workflows
    - **TradingAgents (Tauric)** for agent-based system design patterns
    - **TradingView MCP + Claude integration (Jackson)** for real-time data interaction and workflow integration

All of the above are free and open-source, providing a flexible and cost-efficient foundation for rapid iteration and experimentation.

Compute Infrastructure

Local Machine (Work Laptop)

Recommended:

- MacBook Air (M4 or M5) with high RAM
- OR equivalent Windows laptop with:
    - modern processor (recent Intel / Ryzen)
    - high RAM (16GB minimum, 24GB+ preferred)

Role in System:

- primary development environment
- running data pipelines
- backtesting and simulations (including walk-forward and optimization)
- handling multi-tab, parallel workflows

Rationale:

- High RAM enables parallel workflows and large in-memory operations
- Strong local compute reduces reliance on external infrastructure during early-stage development
- Dedicated work laptop improves:
    - security
    - separation of environments
    - reliability of workflows

Cloud Compute (Optional / Premium Tier)

Recommended Option:

- Runpod (GPU/CPU on-demand instances)
  OR AWS (EC2) for more flexible scaling

Used for:

- large-scale text processing
- batch backtesting across many assets
- scaling simulations and experiments

Rationale:

- Not required for MVP
- Enables scalable compute only when needed
- Keeps fixed costs low while allowing expansion

Note:

- In most cases, a strong local machine is sufficient for early-stage research and testing

---

Rationale

The system is designed to:

- maximize productivity through a unified AI-first workflow (Claude Max)
- maintain a lean and focused toolset
- rely on strong local compute for iteration
- extend to cloud compute only when necessary

This approach balances speed, cost efficiency, and scalability.

TLDR: AI & Compute Stack

Component	Provider	Monthly Cost	Role in System	Alpha Enablement
LLM Platform	Claude Max	$100	Research, coding, agents	Core productivity and signal development engine
Development	VS Code + GitHub	$0	Code + version control	Enables reproducible research
Charting	TradingView Pro	~$50	Real-time analysis + MCP integration	Fast validation of signals
Local Compute	Work Laptop	One-time	Dev + pipelines + simulations	Enables fast iteration and parallel workflows
Cloud Compute (Optional)	Runpod / AWS	Usage Based	Scaling compute workloads	Enables larger experiments when needed

Alternative Data & Research Layer

Purpose

These sources are exploratory and intended to:

- expand the opportunity set of potential signals
- provide non-traditional or forward-looking data
- accelerate idea generation and validation

They are **not required for MVP**, but may provide incremental edge once the core system is validated.

Alternative Data Sources

Prediction Markets Data

Provider: Predmktdata (predmktdata.com)
Pricing: ~$49/month (after free trial)

- Provides historical prediction market data (e.g., Polymarket)
- Captures real-time market-implied probabilities of events

Potential Use Cases:

- macro expectation signals
- event probability tracking (e.g., policy, elections, economic outcomes)
- alternative sentiment layer

C-Suite / Insider Activity (Form 4 Data)

Provider: 13Radar
Pricing:

- Free access to raw data
- ~$12/month for full dataset
- Tracks insider trading activity (Form 4 filings)
- Focus on executive / C-suite transactions

Potential Use Cases:

- management sentiment signals
- alignment between insider behavior and company performance
- exploratory correlation with forward returns

Institutional / AI Research Layer

Provider: 3ai.co
Pricing: Not publicly listed

- AI-focused financial research platform
- Potential complement to existing structured datasets (e.g., LSEG)

Context:

- Shared internally (Aiden) as a potential resource

Potential Use Cases:

- augmenting structured datasets
- exploring AI-driven research workflows

Agent-Native Data

Provider: Valyu

- Valyu API (~$0–$10 usage)
- Used for: retrieving semi-structured data (exec info, sentiment, hiring signals)
- Role: potential enrichment layer for agents
- Status: exploratory, not part of core system

Research & Strategy Discovery Layer

Quantpedia (Prime Subscription)

Pricing: ~$399/year (~$33/month)

- Curated database of 100+ quantitative trading strategies
- Derived from academic research and practitioner studies

Rationale

Quantpedia is included as a **research acceleration tool**, not a direct data source.

It enables:

- rapid access to a large set of validated strategy ideas
- efficient testing of diverse alpha concepts
- reduction in manual research effort

Key Principle:

> The goal of this system is to maximize iteration speed and test a wide range of ideas using modern tooling.

A single validated strategy or adapted signal from this dataset can justify the cost multiple times over.

Positioning

- These strategies are widely known in academic and practitioner circles
- The edge is not in discovery alone, but in:
    - implementation details
    - parameterization
    - regime-specific deployment
    - integration into the broader system

Alternative Data & Research Layer

Component	Provider	Monthly Cost	Role in System	Alpha Enablement
Prediction Markets	Predmktdata	~$49	Event probabilities	Potential macro / sentiment signals
Insider Activity	13Radar	$0–$12	Exec trading behavior	Potential sentiment / alignment signal

AI Research Layer	3ai.co	TBD	AI-driven research	Potential dataset enrichment
Strategy Discovery	Quantpedia	~$33	Research database	Accelerates alpha idea generation

Alpha Summary

System + Alpha Stack (US Growth Equities)

Core System (Engine)

What it does

A systematic long-only growth strategy where:

- **Screener defines the universe**
- **Agents generate structured signals**
- **Ranking selects stocks**
- **Rules execute trades**

Workflow (Simplified)

1) Universe (Screener)

Filter US equities by:

- **liquidity**
- **size**
- **trend**
- **growth**

→ Output: tradable universe

---

2) Agents → Signals

Agents convert raw data into numeric features:

- **Fundamental → growth, margins, revisions**
- **Technical → momentum, trend, extension**
- **Sentiment → tone, concreteness, narrative**

→ **Output: multi-factor feature set**

---

3) Ranking

- **Combine signals into a score**
- **Rank stocks daily**
- **Select top N**

→ **Output: portfolio candidates**

---

4) Strategy

- **Long top-ranked names**
- **Exit via:**
  - **stops**
  - **time**
  - **rank deterioration**

---

5) Portfolio

- **Equal or score-weighted**
- **Risk constraints applied**

Key Principle

> **This is a multi-factor model powered by agent-generated features, not an "AI picking stocks" system.**

The 5 Alpha Ideas (Plug Into This System)

These are feature generators for your agents.

---

1) Infrastructure Constraint Nowcast

What it captures:
AI supply bottlenecks (power, chips, capacity)

Signal:
Constraint severity (text + macro + pricing power)

Edge:
Identifies winners of supply shortages early

---

2) Margin Before Revenue Recognition

What it captures:
AI-driven efficiency showing up in margins first

Signal:
Margin expansion + productivity + confirmed AI deployment

Edge:
Markets underreact to early cost improvements

---

3) Narrative vs Numbers Divergence

What it captures:
Mismatch between story and reality

Signal:
Narrative change − actual performance

Edge:
Short hype, long real performers

---

4) Regime-Conditioned Signal Engine

What it captures:
When signals actually work

Signal:
Volatility, trend, liquidity, macro events

Edge:
Avoids trading in bad environments

---

5) Analyst Revision Diffusion + Tone Gating

What it captures:
Slow-moving analyst updates + confirmation from management language

Signal:
Revision momentum + acceleration + tone filter

Edge:
Improves classic post-earnings drift

How It All Fits Together (This is the real insight)

This is not separate pieces. It is a stacked system:

Step 1: Macro Layer

→ Infrastructure Constraint Nowcast
"What is happening in the real world?"

Step 2: Micro Fundamentals

→ Margin Before Revenue
"Who is actually benefiting?"

Step 3: Behavioral Filter

→ Narrative vs Numbers
"Who is real vs fake?"

Step 4: Confirmation Layer

→ Analyst Revisions
“Is the market catching on?”

Step 5: Timing Layer

→ Regime Conditioning
“When should I act?”

Step 6: Execution Layer

→ Screener + Ranking + Portfolio
“What do I hold?”

April 17th meeting prep & notes

Momentum-focused US equity trading system from AI tailwinds + Regime detection is my research focus

Leopold Aschenbrenner, former open AI researcher, 23, managing 1.5 billion dollar fund, which returned 47% last yr,

Went through his paper called Situational Awareness, 165 page prose on the decade ahead

AI Framework (from paper)

- 3 drivers of progress:
  - Compute scaling
  - Algorithmic improvements
  - “Unhobbling” → models → agents (real work, not chat)

Implication

- AI ≠ tech theme
- AI = full-stack economic shift
- Value spreads across:
  - Infrastructure
  - Algorithms
  - Deployment (agents)

Where value accrues

- **Infrastructure**
  - Compute, data centers, power buildout
- **Algorithms**
  - Efficiency gains ≈ hardware gains
- **Agent layer (most important)**
  - Real workflows replaced / augmented
  - Actual economic impact

Underrated angle

- Security + algorithmic edge underpriced
- Strong algorithms can outweigh compute advantage

Core takeaway

- AI = shifting stack, not single trade
- Edge = tracking where bottleneck moves next

Transition

- "So based on that, I structured my approach around a few core design principles…"

Edge comes from **iteration speed**, not finding one perfect strategy

Focus is on **implementation + deployment**, not just signal discovery

Use **point-in-time data** to avoid survivorship bias and ensure realistic backtests

Core signals: **fundamentals, analyst revisions, momentum, sentiment**

High conviction in **analyst revisions** due to market underreaction

Use **text data (earnings calls, filings)** to capture narrative shifts early

Pair signals with **regime detection** (volatility, macro) for better timing

Start with a **lean data stack** (FMP, Alpha Vantage, SEC) for fast iteration

Use **fast backtesting (vectorized)** first, then validate more rigorously

Build a **continuous feedback loop**: test → evaluate → refine → redeploy

- Key point:
  → All signals are **statistically validated across time and regimes**

Before we get into the tools, I wanted to show you this company, which seems very aligned with what we are aiming to build, finq.ai -> AI ETF US fund -> interesting thing I saw which was aligned with what we are building in the Canadian market, and then if these tailwind factors playout in other markets as well, like for example taiwanese, chinese, south korean, japan (East Asia)

OVERALL OBJECTIVE

The goal is to generate alpha in U.S. growth equities using a systematic, multi-factor approach enhanced by AI/agent-based workflows.

See if there is similar alpha in lesser know canadian equities, and i am very interested to see what extent, the similar tailwinds affect lesser know canadian companies in the same spaces -> mirror the movements, see if it holds in tawiaese, chinese, and south korean as well -> lesser known alpha

The system must:

- Be grounded in proven financial logic (momentum, fundamentals, expectations, sentiment)
- Use modern AI tooling to accelerate signal extraction and iteration
- Be built lean first (MVP), then scaled only after validation

CORE SYSTEM ARCHITECTURE

This is NOT an “AI picks stocks” system.

It is a structured pipeline:

1. Universe (Screener)
- Filter U.S. equities based on:
    - liquidity
    - market cap
    - trend
    - growth
- Output: tradable universe
2. Feature / Signal Generation (Agents)
   Agents convert raw data into structured numeric features:
- Fundamental Agent:
    - revenue growth
    - margins
    - efficiency
    - analyst revisions
- Technical Agent:
    - momentum
    - trend
    - extension
- Sentiment / Narrative Agent:
    - tone (earnings calls, filings)
    - concreteness vs vagueness
    - narrative shifts

Output: multi-factor feature set

3. Ranking Model
- Combine all features into a score
- Rank stocks daily
- Select top N
4. Strategy Layer
- Long top-ranked stocks
- Exit rules:
    - stop loss
    - time-based
    - rank deterioration
5. Portfolio Construction
- Equal-weighted or score-weighted
- Apply risk constraints

ALPHA FRAMEWORK (CORE SIGNAL IDEAS)

The system is built around 5 key alpha ideas:

1. Infrastructure Constraint Nowcast
- Detect supply bottlenecks (AI compute, chips, power)
- Use text + macro + pricing signals
2. Margin Before Revenue Recognition
- Identify companies where efficiency improves before revenue
- Signals: margin expansion + confirmed AI adoption
3. Narrative vs Numbers Divergence
- Compare company story vs actual performance
- Long real performers, avoid hype
4. Regime-Conditioned Signal Engine
- Detect when signals work (volatility, macro, liquidity)
- Avoid trading in bad environments
5. Analyst Revision Diffusion + Tone Gating
- Track analyst estimate changes
- Confirm using management tone from transcripts

Goldreich Anecdote => options hypothesis

DATA STACK (MVP)

Keep lean and cost-efficient.

Market + Fundamentals:

- Financial Modeling Prep (FMP): prices, fundamentals, analyst revisions
- Alpha Vantage / yfinance: prototyping

Filings:

- SEC EDGAR API (Python ingestion → MCP → agents)

Text:

- Earnings transcripts via FMP
- Filings via EDGAR
- News via web + X (e.g., FirstSquawk)

Macro:

- FRED API (rates, inflation, employment)

Screener:

- QuantInvesting (point-in-time, reduces survivorship bias)

AI / TOOLING STACK

- Claude Max → PRIMARY system for:
    - coding
    - research
    - agent logic
- VS Code + GitHub → development
- TradingView (Pro) → real-time charts + MCP integration

Compute:

- Strong local laptop (high RAM)
- Optional cloud (Runpod / AWS) for scaling

ALTERNATIVE / EXPERIMENTAL LAYER

Used only after core signals are validated:

- Prediction markets (Predmktdata)
- Insider trading data (Form 4 via 13Radar)
- Quantpedia (strategy idea discovery)
- Optional: Valyu API (agent-native data enrichment)

IMPORTANT CONSTRAINTS

- DO NOT jump to RL or complex agents first
- DO NOT over-focus on tools
- PRIORITY = validate signals

DEVELOPMENT PHILOSOPHY

1. Start simple
2. Build one working pipeline:
   data → features → ranking → portfolio
3. Validate ONE signal properly
4. Add complexity only after proof

LONG-TERM VISION (NOT MVP)

After signals are validated:

- Introduce agent decision layer
- Add structured evaluation loops
- Potentially use RL (e.g., HUD-style systems) to:
    - improve decision-making
    - optimize execution
    - adapt to regimes

Important:
RL is for improving deployment of signals, NOT creating alpha.

Start by helping me build the FIRST working signal pipeline and backtest it properly.

https://sim.creditus.ca/simulator => Monte Carlo Simulator for statistical validation, and also out of sample walk forward testing

My goal remains to search for alpha in U.S. growth equities through momentum- and sentiment-driven signals. Once we have a working signal pipeline (data, features, ranking), the next step is layering in decision-making agents trained through structured scenarios and evaluation loops.

One area that looks like a strong next layer is agent training and evaluation infrastructure, specifically platforms like HUD.ai (https://www.hud.ai/).

Separately, I also found an open-source framework that could help on the infrastructure side:

https://github.com/GuillermoMalena/avanzai_open

It's a full-stack financial analysis AI platform (FastAPI, agents, frontend) and could be a useful reference for structuring parts of the system, especially around agent workflows and data integration.

"I'm building a momentum-based trading system focused on US growth equities.

Right now, I have a breakout screener that identifies stocks making new highs. The next step is turning that into a real strategy by systematically testing whether those signals actually generate returns.

I'm starting with a simple baseline: buy on breakout and hold for a fixed period, then analyzing performance across time and different market conditions.

From there, I'm layering in a scoring system to separate high-quality breakouts from weak ones, and adding a regime filter so the system only takes trades when momentum is actually working.

Long term, I'm building this into an agent-based system that can adapt factor weightings and improve decision-making through structured evaluation and feedback."

1-Line Positioning (if they interrupt)

"I'm taking a raw breakout screener and turning it into a fully validated, regime-aware momentum trading system with structured decision logic."

Pay -> Other SF AI company 25 to 30 USD,

Italy and Cricket will manage the schedule asynchronously, Europe trip

They said that 22.5 (900 per hour)

Regular interactions that we have -> we are going ot go of of those

First 2 things, bare minimums

Figuring it out, is the key

Distributed and accumulated

35 years doing this

Money price => average price of volume, accumulation, distribution

Limited liquidity
Smaller fund

Laz lobrini
Chuck smth, lseg -> money flow

Merchant bank, capital first company, they are half of portfolio, generate as much wealth as possible clients come for the ride,

Return is the most important thing

Decision Notes For Resources

Market Data: Section 1

- A custom Python-based screener is a long-term goal, but not required for MVP
- Using an existing point-in-time screener allows faster iteration and reduces implementation risk early on
- The screener will also serve as a reference system for understanding how survivorship bias and historical universes are handled
- Over time, key functionality can be replicated internally once requirements are better understood
- FMP was prioritized over Tiingo due to the inclusion of analyst estimates and revisions, which are critical for building additional alpha signals (e.g., revision-based agents)
- Tiingo remains a potential future addition if higher-quality historical price data is required

Fundamentals Data: Section 2

- FMP is reused from Section 1 to avoid duplication of providers and simplify the data stack
- SEC EDGAR is included specifically for point-in-time accuracy and deeper financial context
- The ability to ingest filings via API and route through MCP is critical for enabling agent-based analysis
- Additional providers were excluded to maintain a lean MVP and minimize cost
- Enterprise-level data (e.g., FactSet, Refinitiv) is deferred until alpha signals are validated

Analyst Revisions: Section 3

- Analyst revisions were a primary reason for selecting FMP as the core data provider
- Revision data enables construction of non-obvious alpha signals beyond basic price/fundamental models
- Institutional datasets were excluded due to high cost and unnecessary complexity at MVP stage
- Point-in-time limitations of FMP are acknowledged and will be addressed if signals prove effective
- This section represents a key differentiation layer in the overall system

Text Data: Section 4

FMP is used as the primary source for earnings transcripts to maintain consistency with the broader data pipeline

SEC filings are prioritized for point-in-time accuracy and completeness

News is included as a lightweight, optional signal source, with free aggregation methods sufficient for MVP

LLMs are used strictly as processing tools for extracting structured signals from text

RAG / advanced retrieval systems are intentionally excluded at this stage to keep the system simple and aligned with current engineering capacity

Premium text data providers are deferred until signal value is confirmed

Macroeconomic Data: Section 5

Decision Notes (Internal)

- FRED is sufficient for initial regime modeling
- Focus is on simple, robust macro signals (not overfitted macro models)
- No need for expensive macro datasets at this stage

AI + Compute + Tools

Decision Notes (Internal)

- Claude Max is selected as the central platform to avoid fragmentation across multiple AI tools
- Cursor and ChatGPT are intentionally excluded to maintain focus and streamline workflows
- TradingView is included for real-time validation and integration with agent workflows
- A dedicated work laptop is required for:
    - performance
    - security
    - separation of environments
- Cloud compute is deferred until scaling is necessary
- The stack prioritizes **speed of iteration over infrastructure complexity**

Alternative Data Sources

Decision Notes (Internal)

- Alternative data sources are included as **optional extensions**, not core dependencies
- Prediction markets and insider data are exploratory and require validation before integration
- 3ai.co is included based on internal recommendation and potential synergy with structured datasets
- Quantpedia is prioritized as a **high-leverage research tool**, enabling faster idea generation and testing
- The system's primary goal is to:
    - iterate quickly
    - validate broadly
    - identify a small number of production-grade signals